Teaching Statement

I have taught general-education, core, and applied mathematics to students with widely different levels of preparation and different academic goals. Across these settings, I adapt the pace, structure, assignments, and assessments of each course to the students in front of me.

These experiences have shaped my belief that students should experience each course as one connected mathematical story rather than a collection of facts and procedures. As the course develops, concrete questions lead naturally to abstract ideas and helping students see why each new concept is needed. My goal is for students to build fundamental skills while learning to connect ideas, construct logical arguments, and judge mathematical claims for themselves.

Teaching Across Courses and Student Backgrounds

My teaching has ranged from general-education courses for nonmajors to core and applied mathematics courses. The emphasis changes with the course and with what students need from it. In general-education courses, I make mathematical reasoning accessible to students who may not take another mathematics course. In core courses, I balance computational fluency with the conceptual foundations students will need later, while in applied courses I connect mathematical ideas to problems from STEM. My undergraduate background in engineering helps me understand the mathematical skills students in applied fields need.

Students within the same course also enter with different levels of preparation and mathematical maturity. These differences are especially clear in Analysis and Optimization, which enrolls students from various STEM fields. Although linear algebra is a prerequisite, students enter with widely varying command of it, and many have never written a formal proof. To help them develop proof-writing skills, I redesigned some homework assignments as guided worksheets that provide the structure of an argument while asking students to supply definitions, justifications, and intermediate steps. In course evaluations, students described proofs as more accessible and intuitive and noted that the guided structure helped them understand the material.

I make an effort to learn students' names and majors, encourage them to come to office hours, and offer academic guidance when needed. During class, I invite questions and ask students to anticipate the next step or explain why an argument should work. These exchanges, together with their written work, help me decide when the class needs review, additional examples, or more independence.

Building a Mathematical Story

I organize each course as one connected story in which abstract ideas grow out of concrete questions. Before introducing a theory, I ask what problem motivates it, why the result should be plausible, and what makes it nontrivial. As answers lead to further questions, students see why new ideas are needed and how individual topics form a developing mathematical argument. Throughout the semester, I return to earlier questions so that students can see how their answers become more precise as the theory develops. I also ask students to identify what each method can and cannot answer, since the limitations of one idea often motivate the next. Students then become better able to recognize when familiar methods apply in new settings.

In *Analysis and Optimization*, this story develops through approximation. Students first use linear approximation and the gradient to locate possible extrema. The limits of first-order information then create the need for quadratic approximation and the Hessian to classify local behavior. Gradient descent method shows how these local models can also guide computation. Concrete optimization problems motivate each step and show why the hypotheses of the theory matter.

In *Linear Algebra*, I begin with questions such as why matrix multiplication has its particular definition, how the solvability of $A\mathbf{x} = \mathbf{b}$ changes with $\mathbf{b}$, and how abstract linear transformations can be represented by matrices. Pursuing these questions leads naturally to abstraction, so students encounter general concepts as answers to concrete problems rather than as definitions without motivation. They also develop broader habits of inquiry by asking whether an object exists, whether it is unique, and how it can be classified or represented.

Designing Assessment as Part of Learning

In my current Analysis and Optimization course, I organize homework, quizzes, and exams as parts of a single assessment system. Homework provides low-stakes practice, short quizzes encourage regular preparation, and exams measure independent understanding and reasoning. Quiz questions are selected from assigned homework problems, and quiz performance can count toward the computational part of the corresponding exam. This rewards regular practice with core skills while allowing exams to devote more attention to concepts and proofs. Using several assessments also gives students opportunities to recover from an early setback and demonstrate improvement.

This structure allows me to distinguish among routine computation, conceptual understanding, and proof writing. Mastery of core concepts and computations provides a clear path to success, while theory-based problems allow students seeking greater depth to demonstrate it through explanation and proof. I also choose assessment formats according to what I want students to demonstrate. In Linear Algebra, I used an oral final examination and provided sample problems and possible follow-up questions in advance. This allowed the exam to measure preparation, reasoning, and communication rather than a student's response to an unexpected question.

I revise my assessment design each time I teach a course, using student performance to identify what needs improvement. In my current Analysis and Optimization course, observations from the previous semester led me to introduce more frequent supervised assessments and connect homework, quizzes, and exams more closely. I have also used different approaches to grade recovery. In Calculus, exam corrections encouraged students to revisit their mistakes, while in Linear Algebra, completing review problems before a quiz allowed students to recover part of the points they lost. These experiences guide how I adjust the timing, weighting, and roles of assessments in future courses.

Teaching Mathematics in a Changing Environment

The growing use of AI has made it more important to distinguish between producing a mathematical answer and understanding it. When I discuss AI with students, I emphasize that learning mathematics is essential to using these tools properly and responsibly. This requires more than writing effective prompts. Students need mathematical understanding to examine assumptions, identify gaps in an argument, and decide whether a conclusion follows. Mathematics develops these habits by requiring students to justify each step and verify results for themselves.

As AI-generated solutions have become easier to obtain, I have changed the role of homework in my grading, treating it primarily as practice and relying more on supervised assessments to evaluate independent understanding. I generally require students to submit homework by hand. Even when they have consulted AI or other tools, writing out a solution gives them another opportunity to work through and verify the reasoning before submitting it. I continue to consider how AI can be incorporated into instruction as these tools develop. Technology may change how students produce answers, but my goal is to teach students how to decide whether those answers can be trusted.

Across my courses, my goal is to make critical mathematical thinking accessible to students with different backgrounds. I want students to master fundamental skills, see how mathematical ideas develop from concrete questions, and learn to explain and evaluate arguments for themselves. I will continue to refine my teaching in response to my students and a changing educational environment while bringing this approach to a broad range of undergraduate mathematics courses.